



1. 设 $A \in M_n(\mathbb{C})$, λ 是一个特征值
假设有 3 个关于 λ 的 Jordan 链
 $(A - \lambda I)v_1 = 0, (A - \lambda I)v_2 = v_1, (A - \lambda I)v_3 = v_2$
其中 $m_1 \geq m_2 \geq m_3 \geq 2$, 且
 $(A - \lambda I)v_1 = (A - \lambda I)v_2 = (A - \lambda I)v_3 = 0$
(2) $(A - \lambda I)v_1, (A - \lambda I)v_2, (A - \lambda I)v_3$
线性无关
证明: 3 个 Jordan 链所有向量
线性无关

试举例说明: 若 v_1, v_2, v_3 线性
无关, 3 个 Jordan 链所有向量可能
线性相关.
(本题说明求过渡阵的 Jordan
链时, 只要保证不同 Jordan 链的
链首特征向量线性无关即可)

2. 设 $A, B \in M_n(\mathbb{C}), AB = BA$
(1) 设 $s, t \in \mathbb{C}$, 证明: $e^{As+tB} = e^{As}e^{tB}$
(2) e^A 可逆, $(e^A)^{-1} = e^{-A}$

(3) 证明: $e^A e^B = e^{A+B} e^{AB}$
3. 设 A 是 n 阶方阵, 求
 $\lim_{k \rightarrow +\infty} A^k$ 存在的充要条件及
极限矩阵
4. 设 $A \in M_n(\mathbb{C})$, 求证:
 $\sin 2A = 2 \sin A \cos A$
5. 设 $A \in M_n(\mathbb{C})$, 且 A 可逆.
证明: 存在 $B \in M_n(\mathbb{C}), B^2 = A$.

(B 称为 A 的平方根).
举例说明: 若 A 不可逆, 结论
不正确. $N \in M_n(\mathbb{C})$
6. 设 $A \in M_n(\mathbb{C}), \text{rank } A = 1$
设 $f(z)$ 在 A 的谱上可定义
证明: 存在一个二次多项式 $P(z)$
 $P(z), P(A) = f(A)$
7. 是否存在 $A \in M_2(\mathbb{R}), \cos^2 A = \begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}$ 需要
使得 $\sin A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$? 叫 $\cos A$ 零质
又 2 阶阵 $\cos^2 A = 0_{2 \times 2}$ 矛盾

7. 解: $\begin{pmatrix} 1 & 2020 \\ 0 & 1 \end{pmatrix}$ 不可对角化
 $\Rightarrow A$ 不可对角化
则 $A \sim \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_2 \end{pmatrix} \Rightarrow \sin \lambda_1$ 为 $\sin A$
的特征值
 $\Rightarrow \sin \lambda_1 = 1 \Rightarrow \cos \lambda_1 = 0$ 矛盾
 $\sin A \sim \begin{pmatrix} \sin \lambda_1 & \cos \lambda_1 \\ 0 & \sin \lambda_2 \end{pmatrix}$

Jordan 标准形

设 $A \in M_n(\mathbb{C})$, 最小多项式 $m_A(x) = (x - \mu_1)^{m_1} \cdots (x - \mu_s)^{m_s}$, 次数是 m .
设 $P^{-1}AP = J = \text{diag}(J_{\mu_1, m_1}, \dots, J_{\mu_s, m_s}) = \text{diag}(J_{\mu_1, m_1}, \dots, J_{\mu_s, m_s})$
其中 $J_{\mu_i, m_i} = \begin{pmatrix} \mu_i & 1 & & 0 \\ & \mu_i & \ddots & \\ & & \ddots & 1 \\ & 0 & & \mu_i \end{pmatrix}_{m_i \times m_i}$ (关于特征值 μ_i 的指数 m_i 的 Jordan 块)

函数在矩阵谱上取值

定义 给定函数 $f(z)$, 若数据 $f(\mu_1), f(\mu_2), \dots, f(\mu_s)$ 存在, 则称 $f(z)$ 在 A 的谱上可定义 (defined on the spectrum of A).
定义 给定函数 $f(z)$, 假若 $f(z)$ 在 A 的谱上可定义, 定义 $f(A) = P^{-1}f(PAP^{-1})P = P^{-1}f(J)P = P^{-1}f(\text{diag}(J_{\mu_1, m_1}, \dots, J_{\mu_s, m_s}))P$, 其中
 $f(J_{\mu_i, m_i}) = \begin{pmatrix} f(\mu_i) & f'(\mu_i) & \cdots & \frac{1}{(m_i-1)!} f^{(m_i-1)}(\mu_i) \\ f(\mu_i) & f'(\mu_i) & \cdots & \frac{1}{(m_i-2)!} f^{(m_i-2)}(\mu_i) \\ \vdots & \vdots & \ddots & \vdots \\ f(\mu_i) & f'(\mu_i) & \cdots & f(\mu_i) \end{pmatrix}_{m_i \times m_i}$

函数在矩阵谱上取值

定理 设 $f(z)$ 在 A 的谱上可定义, 则存在唯一次数小于 m 的多项式 $p(z)$ 满足
 $p(\mu_i) = f(\mu_i), i = 1, \dots, s, p(\mu_i) = f(\mu_i), i = 1, \dots, s, p(\mu_i) = f(\mu_i), i = 1, \dots, s$
($p(z)$ 称为 Hermite- Lagrange 插值多项式)
由定理, $p(A) = f(A)$.

1. 证: 设 $\beta_{i,k} = (A - \lambda_0 I)^{m_i-k} v_i, i=1,2,3$
由题设 (1) $(A - \lambda_0 I)^k \beta_{i,k} = 0$
(2) $\beta_{11}, \beta_{21}, \beta_{31}$ 线性无关
断言: $\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}, \beta_{31}, \beta_{32}$ 线性无关
设 $C_1 \beta_{11} + C_2 \beta_{12} + \cdots + C_{12} \beta_{12} = 0$
 $(A - \lambda_0 I)$ 作用
 $\Rightarrow C_{21} \beta_{11} + C_{22} \beta_{21} + C_{12} \beta_{31} = 0$ 成 λ
 $\Rightarrow C_{12} = C_{22} = C_{23} = 0$
 $\Rightarrow C_1 = C_2 = C_3 = 0$

引入符号 $C_v = (A - \lambda_0 I)^t v, (A - \lambda_0 I)^t v, \dots, v$
设 $\tilde{C}_v = \begin{cases} C_v & \text{若 } v = (A - \lambda_0 I)^t v \\ C_v & \text{若 } v \notin C(A - \lambda_0 I) \end{cases}$
假设 $\beta_{11}, \dots, \beta_{1k}, \beta_{21}, \dots, \beta_{2k}, \beta_{31}, \dots, \beta_{3k}$ 线性无关 $\rightarrow C\beta_k$
 $C\beta_k, C\beta_k, C\beta_k$
由断言的方法 $\Rightarrow \tilde{C}\beta_k, \tilde{C}\beta_k, \tilde{C}\beta_k$ 线性无关.

反例: $J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ 取 $v_1 = e_1 + e_2$
 $v_2 = e_2 + e_3$
 $v_3 = e_3$
断言 $Jv_1 = Jv_2 = Jv_3 = e_1$
2. 证: (1) $e^{As} e^{At} = \sum_{j=0}^{\infty} \frac{(As)^j}{j!} \sum_{k=0}^{\infty} \frac{(At)^k}{k!}$
 $= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{s^j t^k}{j! k!} A^{j+k}$
 $= \sum_{j+k=n} \frac{s^j t^{n-j}}{j! (n-j)!} A^n$
 $= \sum_{j=0}^n \frac{s^j t^{n-j}}{j! (n-j)!} \cdot \frac{A^n}{n!}$
 $= \sum_{j=0}^n (s+t)^n \cdot \frac{A^n}{n!} = e^{A(s+t)}$
(2) 取 $s=1, t=-1$
(3) 由 $AB=BA \Rightarrow Ae^B = e^B A$
令 $g(t) = e^{(A+B)t} e^{-Bt} e^{-At}$
 $\frac{dg(t)}{dt} = (A+B)e^{(A+B)t} e^{-Bt} e^{-At}$
 $+ e^{(A+B)t} (-B)e^{-Bt} e^{-At}$
 $+ e^{(A+B)t} e^{-Bt} (-A)e^{-At} = 0_{n \times n}$
 $\Rightarrow g(t) = C$ 常数阵 $C = g(0) = I_n$

3. 证: 设 $P^{-1}AP = J = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{pmatrix}, J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}_{n_i \times n_i}$
 $A^k = P \begin{pmatrix} J_1^k & & \\ & \ddots & \\ & & J_s^k \end{pmatrix} P^{-1}$
只需 $\lim_{k \rightarrow \infty} J_i^k$ 存在 $i=1, \dots, s$ $J_i^k = (\lambda_i I_{n_i} + N_{n_i})^k$
 $= \sum_{j=0}^k C_k^j \lambda_i^{k-j} N_{n_i}^j$
 $\Leftrightarrow \lim_{k \rightarrow \infty} C_k^j \lambda_i^{k-j}$ 存在
 $0 \leq j < n_i$

① $|\lambda_i| < 1 \Rightarrow \lim_{k \rightarrow \infty} J_i^k = 0_{n_i \times n_i}$
② $|\lambda_i| \geq 1$ 且 $\lambda_i \neq 1$ 发散
③ $|\lambda_i| = 1, n_i \geq 2$ (j 可取 1) $\lim_{k \rightarrow \infty} C_k^1$ 发散
④ $|\lambda_i| = 1, n_i = 1 \Rightarrow \lim_{k \rightarrow \infty} \lambda_i^k = 1$
可知 $\lim_{k \rightarrow \infty} A^k$ 存在 $\Leftrightarrow$ ①, ④ $i=1, \dots, s$
极限矩阵为 $P \begin{pmatrix} 1 & \cdots & 0 \\ & \ddots & \\ & & 1 \end{pmatrix} P^{-1}$ (m_i 为代数重数)

5. $P^{-1}AP = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{pmatrix}, J_i = \begin{pmatrix} \mu_i & 1 & & \\ & \mu_i & \ddots & \\ & & \ddots & 1 \\ & & & \mu_i \end{pmatrix}$
 A 可逆 $\Rightarrow \mu_i \neq 0$ 取 $\mu_i \neq 0 \Rightarrow \mu_i^2 = \lambda_i$
考虑 $J_i^* = \begin{pmatrix} \mu_i & 1 & & \\ & \mu_i & \ddots & \\ & & \ddots & 1 \\ & & & \mu_i \end{pmatrix}$ 断言: $(J_i^*)^2 \sim J_i$
 $(J_i^*)^2 = (\mu_i I_{n_i} + N_i)^2 = \lambda_i I_{n_i} + 2\mu_i N_i + N_i^2$
注意到 $\text{rk}((J_i^*)^2 - \lambda_i I_{n_i}) = n_i - 1$
 $\Rightarrow (J_i^*)^2$ 只有一个 Jordan 块 $\mu_i \neq 0$
故 $(J_i^*)^2 \sim J_i \Rightarrow$ 造 B

注: ① 可从构造出 $\tilde{J}_i$ 使 $\tilde{J}_i^2 = J_i$
简单例子:
 $J = \begin{pmatrix} \lambda & 1 & \\ & \lambda & 1 \\ & & \lambda \end{pmatrix}$ 取 $M = I_3 + \frac{1}{\lambda}(\lambda^2 I_3) + \frac{1}{\lambda^2} \cdot \frac{1}{\lambda}(\lambda^2 I_3)^2$
 $= \begin{pmatrix} 1 & \frac{1}{\lambda} & -\frac{1}{\lambda^2} \\ & 1 & \frac{1}{\lambda} \\ & & 1 \end{pmatrix} \lambda \neq 0$

令 $\tilde{J} = \sqrt{\lambda} M$ 验证 $\tilde{J}^2 = J$
② $\forall k \in \mathbb{N}_+, A$ 可逆 则存在 B 使 $B^k = A$
 $J_i = \lambda_i (I_{n_i} + \lambda_i^{-1} N_i)$
回顾 $(1+\lambda)^{\frac{1}{k}}$ 的 Taylor 展开
 $= 1 + \sum_{i=1}^{\infty} \frac{\frac{1}{k}(\frac{1}{k}-1) \cdots (\frac{1}{k}-i+1)}{i!} \lambda^i$
取 $M_i = (I_{n_i} + \lambda_i^{-1} N_i)^{\frac{1}{k}}$
 $= 1 + \frac{1}{k} \lambda_i^{-1} N_i + \cdots + \frac{1}{k} \cdots \frac{(\frac{1}{k}-n_i+2)}{(n_i-1)!} (\lambda_i^{-1} N_i)^{n_i-1}$

$$\begin{aligned} \text{取 } M_i &= (I_{n_i} + \lambda_i^{-1} N_i)^{\frac{1}{k}} \\ &= I_{n_i} + \frac{1}{k} (\lambda_i^{-1} N_i) + \dots + \frac{\frac{1}{k} \dots (\frac{1}{k} - n_i + 2)}{(n_i - 1)!} (\lambda_i^{-1} N_i)^{n_i - 1} \end{aligned}$$

$$\text{则 } M_i^k = I_{n_i} + \lambda_i^{-1} N_i$$

6. 证: $\text{rk } A = 1$ 则 $A = uv^H$ $u, v \in \mathbb{C}^n$

$$\textcircled{1} v^H u \neq 0 \quad A \sim J_1 = \begin{pmatrix} v^H u & & \\ & 0 & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$$

$$\textcircled{2} v^H u = 0 \quad A \sim J_2 = \begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$$

$$\textcircled{1} f(A) \sim f(J_1) = \begin{pmatrix} f(v^H u) & & \\ & f(0) & \\ & & \ddots & \\ & & & f(0) \end{pmatrix}$$

找 $p(z)$ (一次多项式)

$$\text{令 } p(v^H u) = f(v^H u), \quad p(0) = f(0)$$

$$p(z) = \frac{z - v^H u}{0 - v^H u} f(0) + \frac{z - 0}{v^H u - 0} f(v^H u)$$

$\Rightarrow p(z)$ 和 $f(z)$ 在 A 的谱上定义相同

$$\text{则 } p(A) = f(A)$$

(验证 $p(J_1) = f(J_1)$)

$$\textcircled{2} f(A) \sim f(J_2) = \begin{pmatrix} f(0) & f'(0) & & \\ & f(0) & & \\ & & \ddots & \\ & & & f(0) \end{pmatrix}$$

$$\text{找 } p(z) \quad \text{令 } p(0) = f(0) \quad p'(0) = f'(0)$$

$$p(z) = f'(0)z + f(0)$$

$$\Rightarrow p(J_2) = f'(0)J_2 + f(0) = f(J_2)$$

注: $\textcircled{1} m_A(x) = x(x - v^H u)$ 2次 $\rightarrow p(x)$ 1次

$$\textcircled{2} m_A(x) = x^2$$